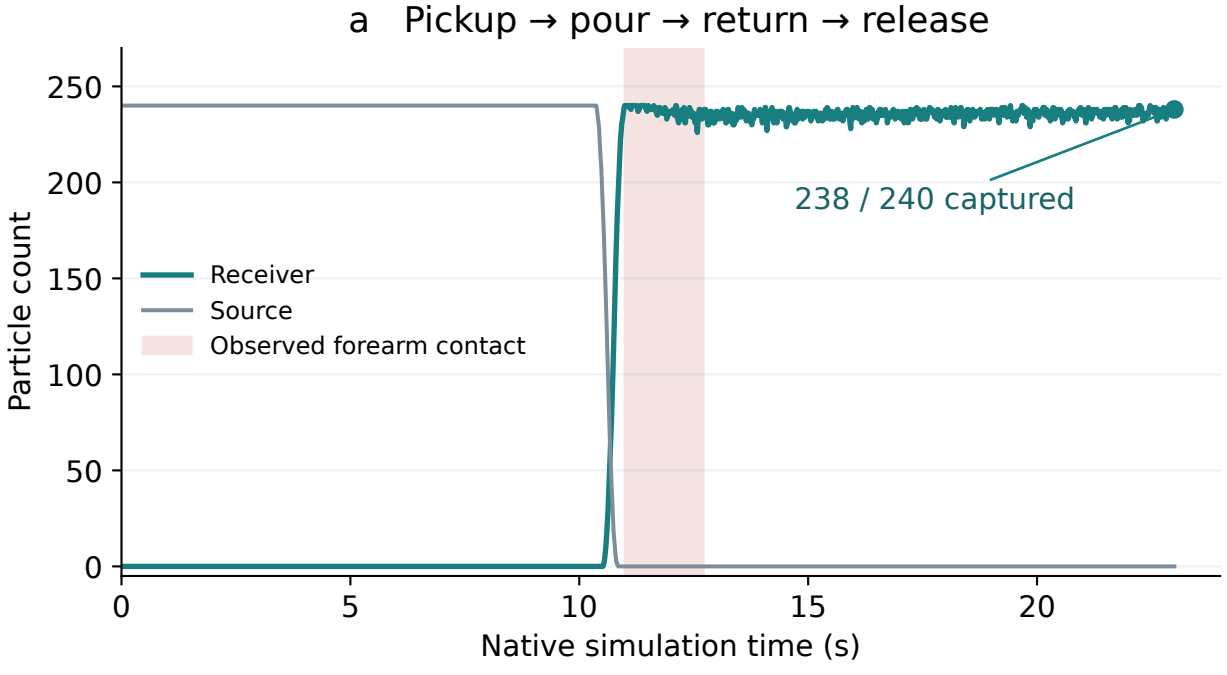
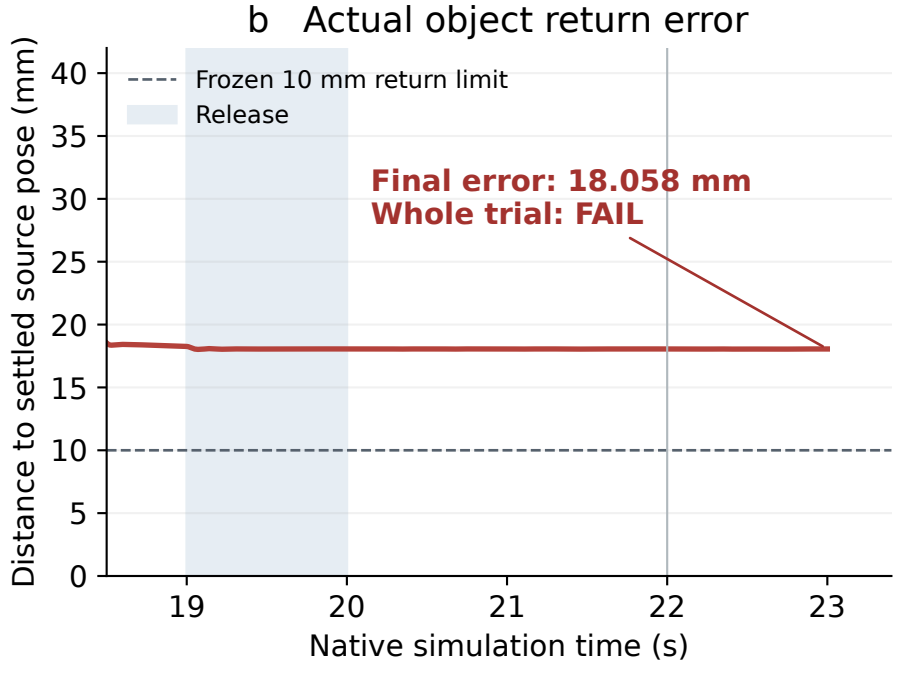
UR5e: 23 s executed; frozen return gate failed

184,000 physics steps · 2,300 control rows · 575 saved native states · independent readback confirmed





13 / 14 frozen gates passed. CFL 0.108; projection convergence 100%; zero particles outside the frozen domain.

Additional quality issue: forearm contact in 172 / 2,300 rows; maximum relative grasp drift 14.764 mm (sampled).

Independent DFSPH with a bench source and scripted actuator control; no original GRID, rack extraction, perception or learned whole-task policy.